

Merewether High School
Year 12 Trial HSC Examination 2001

MATHEMATICS
Extension 2 paper

Time Allowed: 3 hours plus 5 minutes reading time

Instructions:

- All questions may be attempted
- Start each question on a new page
- In every question all necessary working should be shown – full marks may not be awarded for answers without suitable working
- Approved silent calculators may be used
- A Table of Standard Integrals is provided
- Hand in the paper in TWO bundles, Questions 1, 2, 3, 4 and Questions 5, 6, 7, 8.

Question 1

15 Marks

- (a) Evaluate $\int_0^2 \sqrt{4-x^2} dx$ 2
- (b) Find $\int \frac{2x-3}{x^2-4x+5} dx$ 3
- (c) Find $\int \sin^{-1} x dx$ 2
- (d) Evaluate $\int_0^{\frac{\pi}{4}} \sin^2 x \cos^2 x dx$ 2
- (e) (i) Given that $I_n = \int_0^{\frac{\pi}{2}} \cos^n x dx$, prove that $I_n = \left(\frac{n-1}{n}\right) I_{n-2}$,
where n is an integer and $n \geq 2$. 6
- (ii) Hence evaluate $\int_0^{\frac{\pi}{2}} \cos^5 x dx$.

Question 2

15 marks

- (a) If w is the complex number $2\sqrt{3}i - 2$,
(i) find $|w|$;
(ii) find $\arg w$;
(iii) write w in modulus argument form;
(iv) show that $w^2 = 4\bar{w}$. 5
- (b) Sketch the locus described by $\arg(z+1) = \frac{3\pi}{4}$. 2
- (c) Sketch the locus described by $|z-1| = |z+i|$ 2
- (d) Evaluate $\frac{1}{(-1+i\sqrt{3})^6}$. 2
- (e) Find the locus of z if $w = \frac{z-2}{z}$, given that w is purely imaginary. 4

Question 3

15 marks

- (a) By means of the substitution $x = a - u$, or otherwise, prove that 8
- $$\int_0^a f(x) dx = \int_0^a f(a-x) dx.$$
- Hence prove that $\int_0^{\pi} \frac{x \sin x dx}{1 + \cos^2 x} = \int_0^{\pi} \frac{(\pi-x) \sin x dx}{1 + \cos^2 x} = \frac{\pi^2}{4}$
- (b) Sketch the following curves for $-2\pi \leq x \leq 2\pi$.
Do each sketch on a separate diagram.
- (i) $y = |\sin x|$ 2
- (ii) $y = \sin|x|$ 2
- (iii) $y^2 = \sin x$ 3

Question 4

15 marks

- (a) (i) Show that $(1+i)$ is a zero of the polynomial $P(x) = x^3 + x^2 - 4x + 6$. 5
- (ii) Using (i), resolve $P(x)$ into irreducible factors over the field of:
(α) Complex Numbers;
(β) Real Numbers.
- (b) If $x^4 + 2x^3 - 12x^2 + 14x - 5 = 0$ has a root of multiplicity 3,
find all the roots of the equation. 4
- (c) (i) Assuming a, b, c, d are real, show that if the roots of the equation
 $(a^2 + b^2)x^2 + 2(ac + bd)x + c^2 + d^2 = 0$ are real then they are equal. 6
- (ii) Show that the double root in (i) is $x = \frac{-c}{a}$.

Question 5

15 marks

(a) The ellipse E has Cartesian equation $\frac{x^2}{25} + \frac{y^2}{16} = 1$.

(i) Sketch the curve and write down the:

- (α) eccentricity;
- (β) coordinates of the foci S and S' ;
- (γ) equations of the directrices.

(ii) (α) Show that the point P on E can be represented by the coordinates $(5 \cos \theta, 4 \sin \theta)$.

(β) Prove that $PS + PS'$ is independent of the position of P on the curve.

(b) Show that the tangents at the points $P\left(c p, \frac{c}{p}\right)$ and $Q\left(c q, \frac{c}{q}\right)$ on the rectangular hyperbola $xy = c^2$ meet at the point $\left(\frac{2c p q}{p+q}, \frac{2c}{p+q}\right)$.

Question 6

15 marks

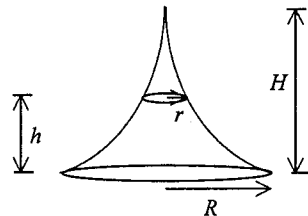
(a) A spire is constructed as shown in the diagram. Each cross-section parallel to the base is a circle

whose radius r is given by $r = \sqrt{R^2 - \frac{R^2 h^2}{H^2}}$

where R is The radius of the base, H is the height of the spire and h is the distance from the base to the circular cross-section.

Prove that the volume of the spire is

$$V = \frac{2}{3} \pi R^2 H.$$



(b) The area enclosed by the curve $y = (x-4)^2$ and the line $y = 16$ is rotated about the y axis. Using cylindrical shells find the volume of the solid generated.

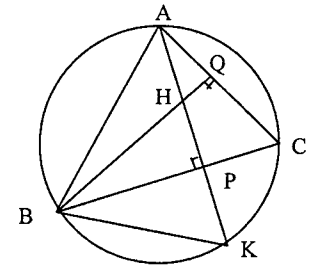
(c) Using the slice technique, find the volume of the solid obtained by revolving $y = \cos x$ about the line $y = -1$ over the interval $-\pi \leq x \leq \pi$.

Question 7

15 Marks

(a) Find $\int \frac{(x^2 + x) dx}{(x-1)(x^2 + 1)}$

(b) The altitudes AP and BQ of an acute angled triangle meet at H. AP produced cuts the circle through A, B and C at K. Prove that $HP = PK$.



(c) (i) The equation $x^3 + 3hx + g = 0$ has two equal roots. Show that this equal root α has the value $\alpha = \sqrt[3]{\frac{g}{2}}$, hence prove that $g^2 + 4h^3 = 0$.

(ii) Using part (i) or otherwise, solve the equation $x^3 - 12x + 16 = 0$ given that it has two equal roots.

(iii) The tangent to the curve $y = x^3$ at $P(2, 8)$ intersects the curve again at Q. Find the coordinates of Q.

Question 8

15 Marks

(a) A farmer uses explosives to blow a stump out of the ground uses enough explosive to hurl debris in all directions with a velocity of 15 ms^{-1} . If the explosion occurs on level ground, show that any person at ground level $10\sqrt{5}$ metres from the explosion could be struck by debris at two instants $\sqrt{5}$ seconds and 2 seconds after the blast, respectively. (Take the acceleration due to gravity as $g = 10 \text{ ms}^{-2}$ and air resistance to be zero)

(b) A curve is defined by the parametric equations $x = \cos^3 \theta$, $y = \sin^3 \theta$ for $0 < \theta < \frac{\pi}{4}$.

(i) Show that the equation of the normal to the curve at the point $P(\cos^3 \phi, \sin^3 \phi)$ is $x \cos \phi - y \sin \phi = \cos 2\phi$.

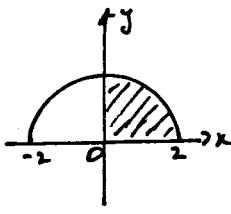
(ii) The normal at P cuts the x axis at A and the y axis at B. Show that $AB = 2 \cot 2\phi$.

Q1(a)

$$\int_0^2 \sqrt{4-x^2} \cdot dx$$

$$= \frac{1}{4} \cdot \pi \cdot 2^2$$

$$= \pi$$



(b) $\int \frac{2x-3}{x^2-4x+5} \cdot dx$

$$= \int \frac{2x-4}{x^2-4x+5} \cdot dx + \int \frac{1}{(x-2)^2+1}$$

$$= \log|x^2-4x+5| + \tan^{-1}(x-2) + C$$

(c) $\int \sin^{-1} x \cdot dx = x \sin^{-1} x - \int \frac{x \cdot dx}{\sqrt{1-x^2}}$

$$= x \sin^{-1} x + \sqrt{1-x^2} + C$$

(d) $\int_0^{\pi/4} \sin^2 x \cos^2 x \cdot dx$

$$= \int_0^{\pi/4} \left(\frac{1}{2} \sin 2x \right)^2 \cdot dx$$

$$= \frac{1}{4} \int_0^{\pi/4} (1 - \cos 4x) \cdot dx$$

$$= \frac{1}{8} [x - \frac{1}{4} \sin 4x]_0^{\pi/4}$$

$$= \frac{1}{8} [\frac{\pi}{4}]$$

$$= \frac{\pi}{32}$$

(e) $I_n = \int_0^{\pi/2} \cos^n x \cdot dx$

$$= \int_0^{\pi/2} \cos x \cdot \cos^{n-1} x \cdot dx$$

$$= [\sin x \cos^{n-1} x]_0^{\pi/2} - \int_0^{\pi/2} \sin x (n-1) \cos^{n-2} x \cdot dx$$

$$= (n-1) \int_0^{\pi/2} \sin^2 x \cos^{n-2} x \cdot dx$$

$$= (n-1) \int_0^{\pi/2} (1 - \cos^2 x) \cos^{n-2} x \cdot dx$$

$$\therefore I_n = (n-1) I_{n-2} - (n-1) I_n$$

$$n I_n = (n-1) I_{n-2}$$

$$I_n = \frac{n-1}{n} I_{n-2}$$

(1) $I_5 = \int_0^{\pi/2} \cos^5 x \cdot dx$

$$\therefore I_5 = \frac{4}{5} I_3$$

$$= \frac{4}{5} \cdot \frac{2}{3} I_1$$

$$= \frac{8}{15} I_1$$

where $I_1 = \int_0^{\pi/2} \cos x \cdot dx = 1$

Q2(a) $w = 2\sqrt{3}i - 2$

$$|w| = \sqrt{(2\sqrt{3})^2 + 2^2} = 4$$

$$\arg w = \tan^{-1}(-\sqrt{3})$$

$$= \pi - \tan^{-1}(\sqrt{3})$$

$$= \pi - \frac{\pi}{3}$$

$$= \frac{2\pi}{3}$$

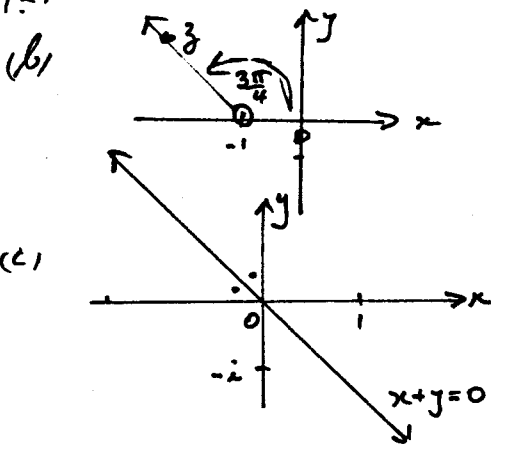
(iii) $w = 4(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3})$

(iv) $w^2 = (2\sqrt{3}i - 2)^2$

$$= -8(1 + \sqrt{3}i)$$

$$= 4(-2 - 2\sqrt{3}i)$$

$$\therefore w^2 = 4w$$



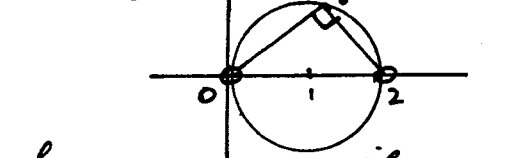
(d) $\frac{1}{(-1+i\sqrt{3})^6} = (-1+i\sqrt{3})^{-6}$

$$= (2 \cos \frac{2\pi}{3})^{-6}$$

$$= 2^{-6} \cos(4\pi)$$

$$= \frac{1}{64}$$

(e) if w is purely imaginary $\arg w = \pm \frac{\pi}{2}$

$$\therefore \arg(z-2) - \arg z = \pm \frac{\pi}{2}$$


Locus of z is the circle $(x-1)^2 + y^2 = 1$ or $|z-1|=1$ excluding $(0,0)$ & $(2,0)$

Q3(a) if $x = a-u$

$$dx = -du$$

$$x=0, u=a$$

$$x=a, u=0$$

$$\therefore \int_0^a f(x) dx = \int_a^0 f(a-u) \cdot (-du)$$

$$= \int_0^a f(a-u) \cdot du$$

$$= \int_0^a f(a-x) \cdot dx$$

$$\int_0^{\pi} \frac{x \sin x \cdot dx}{1 + \cos^2 x} = \int_0^{\pi} \frac{(\pi-x) \sin(\pi-x)}{1 + \cos^2(\pi-x)} \cdot dx$$

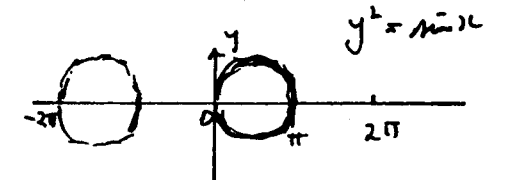
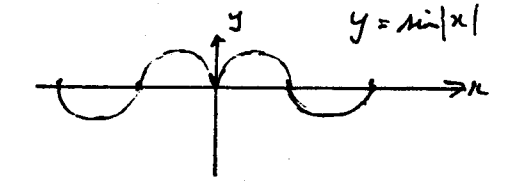
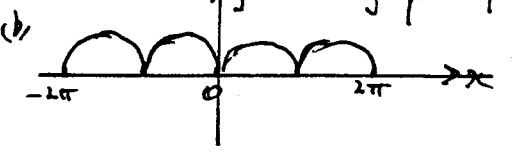
$$= \int_0^{\pi} \frac{(\pi-x) \sin x}{1 + \cos^2 x} \cdot dx$$

$$= \int_0^{\pi} \frac{x \sin x \cdot dx}{1 + \cos^2 x} - \int_0^{\pi} \frac{x \sin x \cdot dx}{1 + \cos^2 x}$$

$$\therefore 2I = \int_0^{\pi} \frac{x \sin x \cdot dx}{1 + \cos^2 x}$$

$$I = -\frac{\pi}{2} [\tan^{-1} \cos x]_0^{\pi}$$

$$= \frac{\pi^2}{4}$$



Q4(a) $P(x) = x^3 + x^2 - 4x + 6$

$$P(1+i) = (1+i)^3 + (1+i)^2 - 4(1+i) + 6$$

$$= 2i(1+i) + 2i - 4 - 4i + 6$$

$$= 0$$

because coeffs. are real $(1-i)$ is also a root

$\therefore x^2 - 2x + 2$ is a factor

(d) $P(x) = (x - (1+i))(x - (1-i))(x+3)$

(b) $P(x) = (x^2 - 2x + 2)(x+3)$

(b) $P(x) = x^4 + 2x^3 - 12x^2 + 14x - 5$

$$P'(x) = 4x^3 + 6x^2 - 24x + 14$$

$$P''(x) = 12x^2 + 12x - 24$$

$$= 12(x-1)(x+2)$$

if $P(x)$ has a triple root it must be $x=1$

$$\therefore P(x) = (x-1)^3(x+5)$$

$\therefore$ roots are $1, 1, 1, -5$

(c) $\Delta = 4(ac+bd)^2 - 4(a^2+b^2)(c^2+d^2)$

$$= -4(ad-bc)^2$$

if roots are real, they must be equal $\therefore ad=bc$

(ii) if roots are equal say λ

$$2\lambda = -\frac{2(ac+bd)}{a^2+b^2}$$

$$\therefore \lambda = -\frac{(ac+bd)}{a^2+b^2}$$

$$= -\frac{(ac + \frac{b^2}{a})}{a^2+b^2} = -\frac{c}{a}$$

$$\frac{x^2}{25} + \frac{y^2}{16} = 1 \rightarrow a=5, b=4$$

$$b^2 = a^2(1-e^2)$$

$$\frac{16}{25} = 1 - e^2, e^2 = \frac{9}{25}$$

$$e = \frac{3}{5}$$

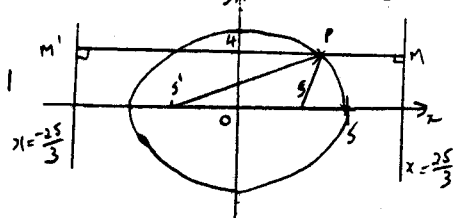
$$S(3,0)$$

$$ae=3$$

$$S'(-3,0)$$

$$\frac{a}{e} = \frac{5}{\frac{3}{5}} = \frac{25}{3}$$

$$\text{Directrices } x = \pm \frac{25}{3}$$



$$P(5\cos\theta, 4\sin\theta): \text{LHS} = \frac{x^2}{25} + \frac{y^2}{16}$$

No LHS $\Rightarrow$
Deriv of Para 1

$$1 = \frac{25\cos^2\theta}{25} + \frac{16\sin^2\theta}{16}$$

$$1 = \cos^2\theta + \sin^2\theta = 1 \text{ c.H.S.}$$

$\therefore P$ lies on E

$$PS = e PM$$

$$PS + PS' = e PM + e PM'$$

$$= e(PM + PM')$$

$$= e MM'$$

$$= e \times 2 \times \frac{a}{e}$$

$$= 2a$$

which is independent of the
Position of P on the curve

$$xy = c^2 \quad P(cp, \frac{c}{p})$$

$$y = \frac{c}{x} \quad Q(cq, \frac{c}{q})$$

$$\frac{dy}{dx} = -\frac{c^2}{x^2}$$

$$\text{Tangent at P: } y - \frac{c}{p} = \frac{-c^2}{c^2 p^2} (x - cp)$$

$$p^2 y - cp = -x + cp$$

$$x + p^2 y = 2cp \quad (1)$$

$$\text{Tan at Q: } x + q^2 y = 2cq \quad (2)$$

$$(1) - (2) \quad (p^2 - q^2)y = 2c(p - q)$$

$$y = \frac{2c(p - q)}{(p^2 - q^2)}$$

$$x = 2cp - \frac{2cp^2}{p + q}$$

$$= \frac{2cp(p + q - p)}{p + q}$$

$$= \frac{2cpq}{p + q}$$

$$= \frac{2cpq}{p + q}$$

$\therefore$ Tangents at P and Q meet at

$$\left(\frac{2cpq}{p + q}, \frac{2c}{p + q} \right)$$

Q6

(a) (5) h



$$r = \sqrt{R^2 - \frac{R^2 h^2}{H^2}}$$

$$A = \pi r^2 = \pi \left(R^2 - \frac{R^2 h^2}{H^2} \right)$$

$$\delta V = \pi \left(R^2 - \frac{R^2 h^2}{H^2} \right) \delta h$$

$$V = \int_0^H \pi \left(R^2 - \frac{R^2 h^2}{H^2} \right) dh$$

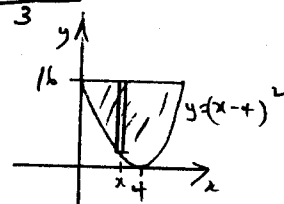
$$= \pi \left[R^2 h - \frac{R^2 h^3}{3H^2} \right]_0^H$$

$$= \pi \left[R^2 H - \frac{R^2 H^3}{3H^2} - 0 \right]$$

$$= \pi R^2 \left(H - \frac{H}{3} \right)$$

$$= \frac{2\pi R^2 H}{3}$$

(b) (5)



$$\delta A = (16 - y) \delta x$$

$$\delta V = 2\pi x (16 - y) \delta x$$

$$V = 2\pi \int_0^8 x (16 - (x-4)^2) dx$$

$$= 2\pi \int_0^8 x (16 - (x^2 - 8x + 16)) dx$$

$$= 2\pi \int_0^8 x (8x - x^2) dx$$

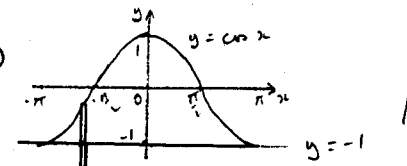
$$= 2\pi \int_0^8 (8x^2 - x^3) dx$$

$$= 2\pi \left[\frac{8x^3}{3} - \frac{x^4}{4} \right]_0^8$$

$$= 2\pi \left[\frac{8 \cdot 8^3}{3} - \frac{8^4}{4} - 0 \right]$$

$$= 2\pi \times 8^4 \times \frac{1}{12} = \frac{2048\pi}{3} \left(682 \frac{2}{3} \pi \right)$$

(c) (5)



$$A = \pi (y+1)^2$$

$$\delta V = \pi (y+1)^2 \delta x$$

$$V = \int_{-\pi}^{\pi} \pi (y+1)^2 dx$$

$$= \pi \int_{-\pi}^{\pi} (\cos x + 1)^2 dx$$

$$= \pi \int_{-\pi}^{\pi} (1 + 2\cos x + \cos^2 x) dx$$

$$= \pi \int_{-\pi}^{\pi} \left(1 + 2\cos x + \frac{1 + \cos 2x}{2} \right) dx$$

$$= \pi \left[\frac{3x}{2} + 2\sin x + \frac{\sin 2x}{4} \right]_{-\pi}^{\pi}$$

$$= \pi \left[\frac{3\pi}{2} + 0 + 0 - \left(-\frac{3\pi}{2} + 0 + 0 \right) \right]$$

$$= 3\pi^2$$

π^2 with incorrect marking - pay 3

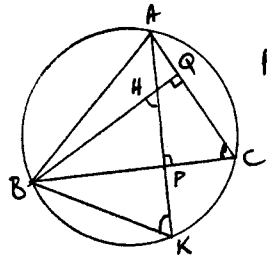
Q7

(a) $\int \frac{x^2+x}{(x-1)(x^2+1)} dx$
 (3)

$1 = \int \left(\frac{1}{x-1} + \frac{1}{x^2+1} \right) dx$

$1 = \ln|x-1| + \tan^{-1}x + C$

(b)
 (4)



Prove
 $HP = PK$

$\widehat{HQC} = \widehat{HPC} = 90^\circ$ (BQ & AP are altitudes)

$\therefore HQCP$ is a cyclic quad.
 (Opp $\angle$ s supplementary)

$\therefore \widehat{BHP} = \widehat{QCP}$ (ext $\angle$ cyclic quad equals int opp $\angle$)

and $\widehat{ACB} = \widehat{AKB}$ (angles in the same segment)

But $\widehat{QCP}$ and $\widehat{AKB}$ are the same angle

$\therefore \widehat{BHK} = \widehat{HKB}$

$\therefore \triangle BHK$ is isosceles (base angles equal)

$BP \perp HK$ (given) is an altitude of $\triangle BHK$

$\therefore BP$ bisects HK (altitude bisects base of isos $\triangle$)

$\therefore HP = PK$

c) $x^3 + 3hx + g = 0$

(4) roots α, α, β .

$\therefore 2\alpha + \beta = 0 \Rightarrow \beta = -2\alpha$

$\alpha^2 \beta = -g$

$\therefore \alpha^2 \times (-2\alpha) = -g$

$2\alpha^3 = g$

$2 \alpha = \sqrt[3]{\frac{g}{2}}$

and $\left(\sqrt[3]{\frac{g}{2}}\right)^3 + 3h \cdot \sqrt[3]{\frac{g}{2}} + g = 0$

$\frac{g}{2} + 3h \cdot \sqrt[3]{\frac{g}{2}} + g = 0$

$\frac{3g}{2} = -3h \cdot \sqrt[3]{\frac{g}{2}}$

$\frac{g^2}{8} = -h^3 \times \frac{g}{2}$

$\frac{g^2}{4} = -h^3$

$2 g^2 + 4h^3 = 0$

(ii) $x^3 - 12x + 16 = 0 \quad \therefore h = -4$
 $\alpha, \alpha, \beta \quad g = 16$

$\therefore \alpha = \sqrt[3]{\frac{16}{2}} = 2$

$\beta = -2 \times 2 = -4$

$\therefore$ Roots are $2, 2, -4$

$\therefore x = 2, -4$

(iii) $y = x^3 \quad P(2, 8)$

(2) $\frac{dy}{dx} = 3x^2$ at $P, \frac{dy}{dx} = 12$

Target at $P: y - 8 = 12(x - 2)$
 $y = 12x - 16$

For $Q, x^3 = 12x - 16$

$x^3 - 12x + 16 = 0$

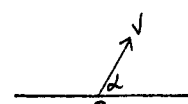
From (i) $x = -4$ for Q

$y = -64$

$\therefore Q(-4, -64)$

$f(x) = x^3 + 3hx + g$
 $f'(x) = 3x^2 + 3h$
 $\alpha = \text{double root}$
 $\therefore 2^3 + 3h \cdot 2 + g = 0$
 $8 + 6h + g = 0$
 $6h + g = -8$
 $h = -\frac{g}{6}$
 $\therefore 2^3 + 3 \cdot \left(-\frac{g}{6}\right) \cdot 2 + g = 0$
 $8 - g + g = 0$
 $8 = 0$
 $\alpha = \sqrt[3]{\frac{g}{2}}$

Q8 (a)
 (7)



$V = 15 \text{ ms}^{-1}$

$g = 10$

$t = 0, x = 0, y = 0$

$\ddot{y} = -g$

$\dot{y} = -gt + C$

$\dot{y} = 15 \sin 2 - 10t$

$y = 15t \sin 2 - 5t^2$

$x = 10\sqrt{5}$, show $t = \sqrt{5}$ or 2

$\dot{x} = 0$

$\dot{x} = V \cos 2$

$x = 15t \cos 2$

$x = 10\sqrt{5}, \frac{10\sqrt{5}}{15} = t \cos 2$

$\cos 2 = \frac{2\sqrt{5}}{3t}$

$y = 0, \text{ find } \frac{5t^2}{15t}$

$\frac{5t}{3}$

But $\sin^2 2 + \cos^2 2 = 1$

$\frac{20}{9t^2} + \frac{t^2}{9} = 1$

$20 + t^4 = 9t^2$

$t^4 - 9t^2 + 20 = 0$

$(t^2 - 4)(t^2 - 5) = 0$

$(t - 2)(t + 2)(t - \sqrt{5})(t + \sqrt{5}) = 0$

$\therefore t = 2, \sqrt{5} \text{ or } -2, -\sqrt{5}$

$t > 0$

$\therefore$ Person $10\sqrt{5} \text{ m}$ from explosion could be struck after $\sqrt{5}$ or 2 seconds

(b) $x = \cos^3 \theta \quad 0 < \theta < \frac{\pi}{4}$

(8) $y = \sin^3 \theta$

(i) $P(\cos^3 \theta, \sin^3 \theta)$

$\frac{da}{d\theta} = 3 \cos^2 \theta \cdot (-\sin \theta)$

$\frac{dy}{d\theta} = 3 \sin^2 \theta \cdot \cos \theta$

$\frac{dy}{dx} = \frac{3 \sin^2 \theta \cos \theta}{-3 \sin \theta \cos^2 \theta}$

$= -\tan \theta$

at $P, \theta = \phi \quad \therefore \frac{dy}{dx} = -\tan \phi$

grad of normal $= \cot \phi$

eqn: $y - \sin^3 \phi = \frac{\cot \phi}{1 - \cos^3 \phi} (x - \cos^3 \phi)$

$y \sin \phi - \sin^4 \phi = x \cos \phi - \cos^4 \phi$
 $x \cos \phi - y \sin \phi = \cos^4 \phi - \sin^4 \phi$

$x \cos \phi - y \sin \phi = (\cos^4 \phi + \sin^4 \phi)(\cos \phi - \sin \phi)$

$x \cos \phi - y \sin \phi = 1 \times \cos 2\phi$

$x \cos \phi - y \sin \phi = \cos 2\phi$

(4) $A(?, 0) \quad B(0, ?) \quad AB = ?$

$y = 0, x \cos \phi = \cos 2\phi$

$x = \frac{\cos 2\phi}{\cos \phi}$

$A\left(\frac{\cos 2\phi}{\cos \phi}, 0\right)$

$x = 0, -y \sin \phi = \cos 2\phi$

$y = \frac{-\cos 2\phi}{\sin \phi}$

$B\left(0, \frac{-\cos 2\phi}{\sin \phi}\right)$

$AB^2 = \frac{\cos^2 2\phi}{\cos^2 \phi} + \frac{\cos^2 2\phi}{\sin^2 \phi}$

$= \frac{\cos^2 2\phi}{\cos^2 \phi \cdot \sin^2 \phi} (\sin^2 \phi + \cos^2 \phi)$

$= \frac{\cos^2 2\phi}{\sin^2 \phi \cos^2 \phi}$

$AB = \frac{\cos 2\phi}{\sin \phi \cos \phi}$

$= \frac{2 \cos 2\phi}{\sin 2\phi}$

$= 2 \cot 2\phi$

(a) $\rightarrow \sin 2\theta = \frac{4\sqrt{5}}{9}$

$\therefore \cos 2\theta = \frac{1}{9}$

$\cos^2 \theta = \frac{1 + \cos 2\theta}{2} = \frac{5}{9} \quad \therefore \cos \theta = \frac{\sqrt{5}}{3}$

$\sin^2 \theta = \frac{1 - \cos 2\theta}{2} = \frac{4}{9} \quad \therefore \sin \theta = \frac{2}{3}$

etc: